# Early Projection and Polynomial Centers

Ten positive unknowns: 97/202, 111/198 and 119/190

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## Quotient centers and early projection in the free ten-positive chart

### Result, domain, and attribution

This package gives three ordinary polynomials in exactly ten independently quantified strictly positive integer variables:

| Graph prefix | Operations | Exact degree | Transport | Root scale |
|---|---|---|---|---|
| quotient_t | 97 | 202 | bV | 2 |
| product2 | 111 | 198 | bV | 2 |
| product4 | 119 | 190 | b^2 | 4 |

Every graph is supplied at two separate complete native/refutation calibrations: historical indexed943/radix67 and current low-padding2963. The latter has 69,685 raw coefficient bytes, not the frozen4177 tuple's 98,219 bytes. Each of its four integers, native proof, exact942-selector bijection, physical marker and Pclp controller changes as one coherent unit. No coefficient port, old word or controller certificate is transferred alone.

The free chart Y=s, lambda=q, inverse-Lucas correction and unsquared trace are credited to source172, whose full retained parent is `parent/`. Source178 supplies the A-quotient idea and the full-product Omega4 evaluator; its original proof and algebra are retained in `inherited/`. Those incoming scaled-chart profiles 96/220 and116/208 do not improve the prior free-chart 90/206. Our contributions are the compatible free-chart composition, the cheaper t=A-1 quotient, first-product truncation, and early p=3 projector exclusion which newly licenses square transport here. The last step avoids illicitly importing the scaled-chart q|Y argument.

All binary +,-,* gates, squares and scalar products are charged. Fixed integer leaves and reuse are free. These are measured tradeoffs, not a global optimality or literature-priority claim. Ordinary soundness/completeness, composed with the identified retained syntactic calibration, proves in ZFC:

```
The equation has no positive-integer zero iff Con(ZFC).
```

No consistency decision, actual zero, unsolvability assertion, new compiled formal-kernel replay or new encoded internal-ZFC proof object is claimed. Finite algebra and byte checks do not prove the unbounded lemmas by testing.

### 1. The complete ordinary polynomials

The independent variables in graph order are

```
J,F,alpha,i,T,w,s,n,eta,zeta > 0.
```

Fix the entire chosen tuple beta,K0,MC,MF. Both actual tuples satisfy beta>=73 odd, K0>0 and 0<M-C<beta<MF<2beta. The strengthened beta bound is used explicitly in section3; no theorem for arbitrary beta>=31 is asserted. MF is the source mask with beta added exactly once, not the native mask. Define these polynomial abbreviations, none a new quantified variable:

```
h=beta J, q=h+1, C=q-F-alpha, X=qw, Y=s, Ybar=qY,
k=eta+zeta, c=kY+eta,
A=Ybar(X+1)+1, Delta=A^2-1, H=4A-5,
v=X Ybar^2, ell=vk, E=1+ell(ell+k),
```

```
R=[h(q-F)+alpha](q^2-1)+(MC+qMF)J,
B=(K0+w)C-F, U=ic^2, a=Delta U^2E, F0=1+a,
G=EF0, f=a+F0, Q=4Delta aF0,
b=cT-R, V=fb-c, K=X(3c-2X)+2,
e6=6(k-R-1)-ell(k^2-4), D=6q^7, M=hDH.
```

Choose a row below; the formulas are division-free integer polynomials. For quotient_t put

```
Qt=(kYbar)[ell+k-(kYbar)Ybar],
C2=U(2AE-Qt), Omega=C2 V, r=2, L=bV.
```

For product2 put

```
S=(kYbar)[ell+k-(kYbar)(Ybar+1)], theta=U^2b,
Omega=U{2A[EV-theta(E^2+S^2)]
              +S[4theta E-(b-c)]}, r=2, L=bV.
```

For product4 put

```
S=(kYbar)[ell+k-(kYbar)(Ybar+1)], theta=U^2b,
e2=(kYbar)^2, Rc=theta(3E+e2)-(b-c),
Sc=4E^2+3S^2+e2^2,
Omega=U[2S Rc+A(4EV-theta Sc)], r=4, L=b^2.
```

In every case set

```
Z0=hD[H(n-LB)+K]-H[DB+h(e6-6q^4s)],
Z=Z0-M Omega,
P=Q(Z^2-r^2M^2GV^2)+(r^2M^2G-Z^2).                    (1)
```

The single equation is P=0. No rational constant, division, recurrence, radical, floor, variable exponent, hidden witness or oracle is a graph gate. The quotient_t implementation shares V in n-V(bB+C2) and computes 2A=(A-1)+(A+1) using two already-live registers.

For proof only put t0=B/h, kappa=K/H, rho=(e6/6)/q^7, sigma=Y/q^3, W=1+hL and z=Z/M. Before integrality is established these are rational quantities. Identically,

```
z=n+kappa+sigma-Wt0-rho-Omega,
P=(Q-1)Z^2-r^2M^2G(QV^2-1).                        (2)
```

## 2. Root extraction before sign or denominator assumptions

At every positive input q>=74, X>=q, Y>=1, c>=3, A>q^2, H>0, U>=c^2, f>2c and Q>c^2. The exact identities

```
f^2-1=4aF0, Q=Delta(f^2-1)=4Delta^2U^2G,
gcd(E,F0)=1, f-1=2Delta i^2c^4E
```

are unchanged from the retained parent. V cannot vanish. If P=0, the integer equation(2), valuations at primes dividing G, and the square factor r^2 show that G is a square. Thus E=tau^2, F0=f0^2 and g=tau f0>0. The parent's rational adjacent-Pell descent, unchanged by the constant square scale, gives y=|z|/(rg) an integer and

```
QV^2-(Q-1)y^2=1, z=epsilon rgy, epsilon=+/-1.
```

In particular z is integral. The signed all-index folding theorem applies with I=2Ug, f^2-Delta I^2=1, Q=(Delta I)^2 and c^2|I. It gives c=psi_A(p) for p positive odd; c>=3 excludes p=1. Hence

```
p>=3, c>=c3=4Delta+3>4q^4, K>0.
```

All of this is independent of the signs of R,B,C,b,V. No q|Y, authentication p=R, auxiliary-index-three exclusion or large-rank ratio bound has been used.

The first square gives k=2psi_(1+2v)(m), m>=1. m=1 gives k=2, eta=zeta=1 and c=2Y+1 strictly between the first two main ordinates, so m>=2. The first base 1+2v>A, and c>k. Strict base/index monotonicity therefore gives m<p: if m>=p, psi_(1+2v)(m)>psi_A(p)=c, impossible. This m<p fact does NOT use p=R or any mask bound.

## 3. New early projector exclusion of main index three

Integrality of z already gives H|Z. Reducing the literal numerator(1) modulo H gives H|hDK. Since D=6qˆ7 and gcd(H,q)=1,

```
H | 6hK.                                    (3)
```

The shift M Omega and either transport are multiples of H; they do not change this step. We retain the entire factor6h.

Suppose p=3. Then m=2 by section2. Write u=X+2-2Y. Using c=4Aˆ2-1, k=8Xqˆ2Yˆ2+4, eta=c-kY gives the exact polynomial identity

```
eta=4qˆ2Yˆ2(Xu+1)+8qY(X+1)-4Y+3.           (4)
```

If u<=-1, the first term is at most -4qˆ2Yˆ2(X-1). Since qY>=32, X>=32, we have qY(X-1)>2(X+1)+1, so(4) is negative. If u>=2, subtracting k from(4) leaves at least 4qˆ2Yˆ2+8qY(X+1)-4Y-1>0. Both alternatives contradict 0<eta<k. Thus u is0 or1, so 2Y=X+2 or X+1 and

```
H=4qY(X+1)-1>2qXˆ2.                        (5)
```

For p=3 the projector is the direct identity

```
4K+(X-8)(8X+1)=3XH(4A+5).                  (6)
```

Together(3),(6) show H divides the positive integer N=6(q-1)(X-8)(8X+1). Because X>=32, 0<N<48qXˆ2. By(5), j=N/H is an integer with1<=j<=23. Modulo q, H congruent -1, X congruent 0 and N congruent 48, so j congruent -48 modq. But49<=j+48<=71<q, impossible. Therefore p>=5 and c>c3. This is a universal proof at the explicitly verified beta>=73 boundary; finite tests of small inputs are not its justification.

Now b cannot vanish by the adjacent-square gap argument from the parent. It has the sign of V. The auxiliary Pell index is odd and at least3. At index3 its two signs would give respectively

```
f(b-4Delta f)=c-c3,
f(b+4Delta f)=c+c3.
```

Both right sides lie strictly between0 and2c<f, impossible. Thus that index is at least5, and

```
|V|>=16Qˆ2-20Q+5>8Qˆ2,
|b|>8fˆ3-1/2>7fˆ3, f>6g,
y<3|V|/2, |V|<3f|b|/2.                     (7)
```

This finally licenses square transport. The proof did not silently exclude an index-three branch before its required early projector argument. For the two bV rows the weaker inherited bound |b|>4Delta f-1 would already suffice: Delta>=4 and f>6g imply 4Delta f-1>96g-1>72g for integer g>=1. their validity does not depend on square transport.

## 4. Centers are small, with both signs covered

Let C0=AUE. Since G=E+Delta Uˆ2Eˆ2,

```
0<C0<3g/2.                                 (8)
```

For quotient_t set Rt=1+kˆ2Ybarˆ2(Ybarˆ2-1)>0. Exact polynomial division by t=A-1 gives E=(A-1)Qt+Rt. Also Qt=kˆ2Ybar[(X-1)Ybarˆ2+1]>0, hence Qt<E/(A-1). Consequently 0<C2=U(2AE-Qt)<2AUE. Thus Omega has the sign of V and

```
2gy+|Omega|<3g|V|+3g|V|=6g|V|<L/2.         (9)
```

For the product rows, temporarily regard A as an indeterminate T over Z[Ybar,k,U,b,c]. Then

```
E(T)=e2 T^2+e1 T+e0,
e2=k^2Ybar^2,
e1=k^2Ybar(1-2Ybar-2Ybar^2),
e0=1+k^2Ybar(Ybar+1)[Ybar(Ybar+1)-1],
V(T)=(1+2(T^2-1)U^2E(T))b-c, F(T)=E(T)V(T).
```

Let Qj be the value at A of the polynomial quotient of F(T) by T^j. Then exactly

```
Omega2=U(2AF-Q1),
Omega4=U(4AF-2Q1-Q3/2).                    (10)
```

Q3 is coefficientwise even, because the only odd-coefficient candidate (b-c)E has T-degree at most2. Equations(10) are proof-only descriptions; the literal evaluators in section1 use no quotient or division gate.

For a polynomial, superscript# denotes absolute-coefficient evaluation at A. Put delta=(Ybar+1)/A<1/16. The coefficient bounds |e1|<2e2(Ybar+1), 0<e0<=e2(Ybar+1)^2 give E#(A)<2E. Since c+1<2U^2E and |V|>(f-c)|b|>Delta U^2E|b|,

```
F#(A)<2E(|b|+c)+8(A^2+1)U^2E^2|b|,
F#(A)/|EV|<8+20/Delta<10.
```

Deletion of low coefficients gives |Qj|<=F#(A)/A^j. As A>=16,

```
|Omega2-2AUEV|<AUE|V|,
|Omega4-4AUEV|<AUE|V|.
```

Therefore both have the sign of V, and

```
|Omega2|<3AUE|V|,
3AUE|V|<|Omega4|<5AUE|V|.                  (11)
```

For product2 with L=bV, (8),(11) and |b|>72g yield 2gy+|Omega2|<(15/2)g|V|<L/2. For product4 with L=b^2, (7),(8),(11) give

```
4gy+|Omega4|<(9/4)f|V|
  <(27/8)f^2|b|<b^2/4.                      (12)
```

The last inequality follows from |b|>7f^3 and f>2. All bounds precede sign recovery and cover negative b,V. We do not call Omega positive until V itself has been proved positive.

## 5. Signed bootstrap, authentication, and projection order

The inverse correction satisfies e*=e6/6<=k-R-1<b for every integer R, so rho<|b|/q^7. In both weights L>0 and L>=|b|. If B<=-1, Wt0=B/h+LB<-L. Solving(2) for positive n and using(9)–(12), positive kappa,sigma, and q^7>2 gives

```
n< L/2-L+|b|/q^7<0,
```

a contradiction. Thus B>=0, and (K0+w)C=B+F>0 gives C>0. Only now use the selected whole tuple's rank bound3q<R<q^4-2. As c>4q^4>2R, b,V are positive. The unchanged normalized main-index and signed auxiliary-index theorems applied to V congruent -c modf and V congruent -R modc identify p=R. Mask parity now gives q even,J odd.

Every Omega is an integer polynomial. Its numerator contribution M Omega is divisible by D and H, so all subsequent modular reductions are exactly those of parent/PROOF.md sections4–5. We state their noncircular order. First k=2psi_(1+2v)(m) is even and e* is integral. Literal cancellation of6 in the rational carrier gives q^4|e*. The full Lucas identity is

```
k=2m+8v Cm+32v^2Zm, Cm=binom(m+1,3), Zm>=0,
3[e*-(2m-R-1)]=4v^2[24Zm-Am Sm],
Am=Cm+4vZm, Sm=k^2+k(2m)+(2m)^2-4.
```

With v=qˆ3wsˆ2 the latter gives qˆ4|e*-(2m-R-1) at every prime, including3. Since m<R and R<qˆ4, m=(R+1)/2. One does not yet infer qˆ7|e*. Write R=2r0+1; r0>=48. The lambda-uniform ratio theorem at lambda=q,Y>=1 gives Y>Xˆr0/3 and Xˆ(r0+1)<H. The full-product projector bound, retaining6h, gives X=2ˆR. Hence q is a power of2; rank growth supplies qˆ6|X and v is divisible by qˆ8. Only now does the cubic identity imply qˆ7|e*. Thus rho and kappa are integral.

The remaining carrier relation yields qˆ3|Y and h|B, since W congruent 1 modh. Therefore t0 is a nonnegative integer. The retained fixed-minus parity, lambda-uniform half-binomial rounding and fresh native reconstruction at A_source=Y(X+1)+2 produce actual source words for the selected whole calibration. This base is not confused with A=qY(X+1)+1. The native interpretation yields a finite syntactic ZFC refutation, not merely a conclusion from elementary scalar boxes.

## 6. Constructive strictly positive completeness

Conversely, the chosen whole native/refutation calibration supplies fresh canonical data q=2ˆa, X=2ˆR, R congruent 3 mod4,3q+1<=R<qˆ4,qˆ3|Y,C>0 and t0=B/h>=0 integral. Parent section6 is uniform in these data and constructs the free-chart inputs w=X/q,s=Y, positive eta,zeta,i,T, with c=psi_A(R), k=2psi_(1+2v)((R+1)/2), g>0,V>0,y>0 and

```
g>kappa+sigma+|rho|.
```

No old packed word or strong period is reused across calibrations. Every center is positive when V>0 by section4. Set, at the selected transport and root scale,

```
n=rgy+Omega+(1+hL)t0+rho-kappa-sigma.        (13)
```

All terms are integers and the preceding strict bound makes n>0. The literal numerator satisfies z=rgy, so(1) vanishes. This construction also handles possibly negative rho. For bV one may equivalently take n_new=n_parent+(r-1)gy+Omega. For bˆ2 we use(13), not a claim that subtracting bVB from an arbitrary positive parent n preserves positivity. All ten independent coordinates have their declared strict positive domain.

## 7. Entire low-padding2963 native calibration

The new native proof, decoder, language and every historical ordinary interface it names are retained in native2963/, with267 exact source pins. Its complete proof is research/low-padding-independent/PROOF.md. The admissible selector bijection g(0)=0,g(j)=j+1 (1<=j<=941) removes exactly old forbidden selector1. It preserves and reflects the full language, including the complete436-character seed,430 actions, renamed initializer928 and calibrated final tile941; it is not an unproved canonical subfamily.

There are3297 coefficient slots,976 weighted positions and978 physical positions. Physical dummy977 and marker1005 have zero coefficient but remain Boolean positions. A separate kernel monomial copies Start0 to1005. All916 padding tests use bit2 of fields1..916. The sole clause is1956; marker parity, Boolean temporal count, alignment, carries and reset tests are proved before the resulting native interpretation. Soundness needs no controller premise. Completeness uses the same tuple's discharged Pclp:

```
rho_native=2^47, L_native=2963, d=139261,
fixed prime factors47,2963, orders23,2962,
spacing68126, auxiliary prime1362521,
H residues29,2938,492699.
```

The finite checker reconstructs the exact four unsigned big-endian integers, not just their boxes; MF_source adds beta once. Every raw/compressed hash, bit length, minimal encoding, marker monomial and physical support is checked. The complete native section12 constructs fresh canonical outer data matching section6 exactly. The free-chart endpoint in section5 constructs fresh original source witnesses, so the native sections5–8 apply. This establishes both interfaces rather than substituting a new tuple into an old calibration.

The historical radix67 specialization is separately supplied with the full parent finite recipe and little-endian payloads,295388 raw bytes. The new tuple is a different specialized polynomial and is not claimed identical to either radix67 or any older4177/4027 specialization.

## 8. Exact algebra, degree, and complete highest forms

The checker independently expands the center over Z[A,Ybar,k,U,b,c], using weights(4,2,1,5,4,2). It compares the division-free actual graph against(10) and the t-quotient identity. Put Def=Omegaˆ2-rˆ2GVˆ2. The weighted degrees of Omega and Def are

```
quotient_t:59,106; product2:59,102; product4:59,94.
```

For quotient_t the exact root defect is

```
C2ˆ2-4G=Uˆ2[4E(Rt-Qt)+Qtˆ2]-4E.
```

No positivity of Rt-Qt is required. The full three defect leaders are

```
16Aˆ10 Uˆ6 Ybarˆ10 bˆ2 kˆ8-16Aˆ11 Uˆ6 Ybarˆ8 bˆ2 kˆ8,
4Aˆ10 Uˆ6 Ybarˆ8 bˆ2 kˆ8,
8Aˆ8 Uˆ6 Ybarˆ8 bˆ2 kˆ8-48Aˆ6 Uˆ6 Ybarˆ12 bˆ2 kˆ8.
```

The complete output identity is

```
P=Q[Z0ˆ2-2M Z0 Omega+Mˆ2Def]+rˆ2Mˆ2G-(Z0-M Omega)ˆ2.
```

Independent degrees are deg(Q,M,G,V)=(72,12,46,36). For bV, degZ0=54; for bˆ2, degZ0=22. The corresponding summand bounds are(180,197,202,70,142),(180,197,198,70,142), and (116,165,190,70,142). In each case the defect summand uniquely dominates. Let q0=beta J,k0=eta+zeta. The complete collected top forms are

```
[P_quotient_t]_202 = 36864 q0ˆ80 wˆ22 sˆ55 iˆ10 k0ˆ32
                           (q0-F)ˆ2(s-w),
[P_product2]_198 = 9216 q0ˆ78 wˆ22 sˆ54 iˆ10 k0ˆ32(q0-F)ˆ2,
[P_product4]_190 = 18432 q0ˆ74 wˆ18 sˆ52 iˆ10 k0ˆ32
                           (q0-F)ˆ2(wˆ2-6sˆ2).
```

They have198,99,198 terms respectively and are nonzero for each actual positive beta. All terms, not a selected numerical ray, are independently collected. Every gate is covered by an exact free-ring cut, all registers, inputs and ports are live, and both alternative operators at every gate are rejected:194,222,238 corruptions per tuple. Normal and optimized Python run the same checks, propagate their mode to child replays, and produce equal receipts. Relocated replay uses only this package.

# A Complete historical coefficient recipe

In this appendix $K$ denotes the port $K_0$ in the new equations. Every symbol introduced here is a fixed constant, independent of the ten quantified variables. The formulas and 436-character seed are the retained indexed BFS943 radix67 specification; they reconstruct the exact supplied payloads, with $\beta = 2^{590069} - 1$.

**The four fixed integers.** Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) = {} & (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ & + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} = {} & \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ & (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ & (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

**The 943 indexed fixed tiles.** Use the actions $u_j, v_j, d_j$ decoded below. Let

$$\begin{gathered}h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\ \mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\ r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.\end{gathered}$$

An interval ending at $-1$ is empty. The table defines $t_i$ for $0 \le i \le 942$; all index ranges are fixed:

| index $i$ | tile $t_i = (w, n, e, s)$ |
|---|---|
| $0, 1$ | $(2, 435, 3, 1),\ (5, 0, 0, 0)$ |
| $j + 2\ (j = 2q + b < 430)$ | $((1 - d_j) l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$ |
| $432 + b\ (b = 0, 1)$ | $(0, h(215, b), 0, h(215, b))$ |
| $434 + b\ (b = 0, 1)$ | $(0, b, 0, b)$ |
| $424 + 2r_q + b\ (q \in \mathcal{R}, b = 0, 1)$ | $(r_q, b, 0, h(q, b))$ |
| $424 + 2l_q + b\ (q \in \mathcal{L}, b = 0, 1)$ | $(0, b, l_q, h(q, b))$ |
| $924 + j\ (0 \le j < 4)$ | $(0, j + 430\lfloor j/2 \rfloor, 0, 438)$ |
| $928$ | $(2, 434, 2, 0)$ |
| $929$ | $(3, 436, 4, 4)$ |
| $930$ | $(4, 436, 4, 0)$ |
| $497 + t\ (t = 434, 435, 436)$ | $(1, 438, 1, t)$ |
| $934 + 3a + b - 6\lfloor a/4 \rfloor$ $a \in \{0, 1, 4\},\ b = 0, 1, 2$ | $(a, 437, b, 437)$ |

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

**A closed decoder for the 430 actions.** Put $D_b(A, k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \lfloor \mathcal{A}/2^{636} \rfloor \bmod 2^{1675},$$
$$E_0 = \lfloor \mathcal{A}/2^{2311} \rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

**The integer $\mathcal{A}$.** The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq2OUTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
OOe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

# B Complete low-padding2963 coefficient recipe

Here $K$ denotes the fixed port $K_0$. No symbol in this appendix is another unknown. **The four fixed integers.** Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{2963}, \quad \beta = B - 1,$$
$$G_m(x) = \sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1956}.$$

$$\begin{aligned} M_C &= \beta - \rho^{977} \\ &\quad - G_{942}(\rho) - \Psi(\rho, 1), \\ M_F &= \beta + 4G_{917}(\rho) + (2^{44} - 2)\Lambda \\ &\quad + \rho^{960} G_9(\rho) + \rho^{987} G_8(\rho) + \rho^{1005}, \\ K &= \rho^{18} + \rho^{1005} + \rho^{2972} \\ &\quad + \Lambda\Phi(1/\rho), \end{aligned}$$

where

$$\begin{aligned} \Phi(x) &= \sum_{i=0}^{941} [1 + 2^{10}(1 + w_i + 2^8 n_i \\ &\quad + 2^{17} p_i + 2^{25} s_i)] x^i + 2^{10}\Psi(x, 2), \\ \Psi(x, t) &= x^{969}(x^9 + t^{17}) G_8(tx) \\ &\quad + t^8 x^{942}(x^{18} + t^{17}) G_9(tx) \\ &\quad + (2 - t) x^{1005}. \end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

**The 942 indexed fixed tiles.** Use the actions $u_j, v_j, d_j$ decoded below. For $d = 0, 1$, let

$$h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q - 1\}|.$$

An interval ending at $-1$ is empty. The table defines $t_i$ for $0 \le i \le 941$; all index ranges are fixed:

| index $i$ | tile $t_i = (w_i, n_i, p_i, s_i)$ |
|---|---|
| $0$ | $(2, 435, 3, 1)$ |
| $j + 1$ $(j = 2q + b < 430)$ | $((1 - d_j) r_{0,u_j}, h(q, b), d_j r_{1,u_j}, v_j)$ |
| $433 - 2a + b$ $(a, b = 0, 1)$ | $(0, 432a + b, 0, 432a + b)$ |
| $\substack{423 + 2r_{d,q} + b \\ q \in \mathcal{S}_d,\ d,b = 0,1}$ | $(d r_{d,q}, b, (1 - d) r_{d,q}, h(q, b))$ |
| $923 + 2a + b$ $(a, b = 0, 1)$ | $(0, 432a + b, 0, 438)$ |
| $927$ | $(2, 434, 2, 0)$ |
| $928$ | $(3, 436, 4, 4)$ |
| $929$ | $(4, 436, 4, 0)$ |
| $496 + t$ $(t = 434, 435, 436)$ | $(1, 438, 1, t)$ |
| $933 + 3a + b$ $(a, b = 0, 1, 2)$ | $(a^2, 437, b, 437)$ |

**A closed decoder for the 430 actions.** Put $D_b(A, k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For

$0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(E_0, \tau_q) - 14),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j)D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

**The integer $\mathcal{A}$.** The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq2OUTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSD0Tgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```